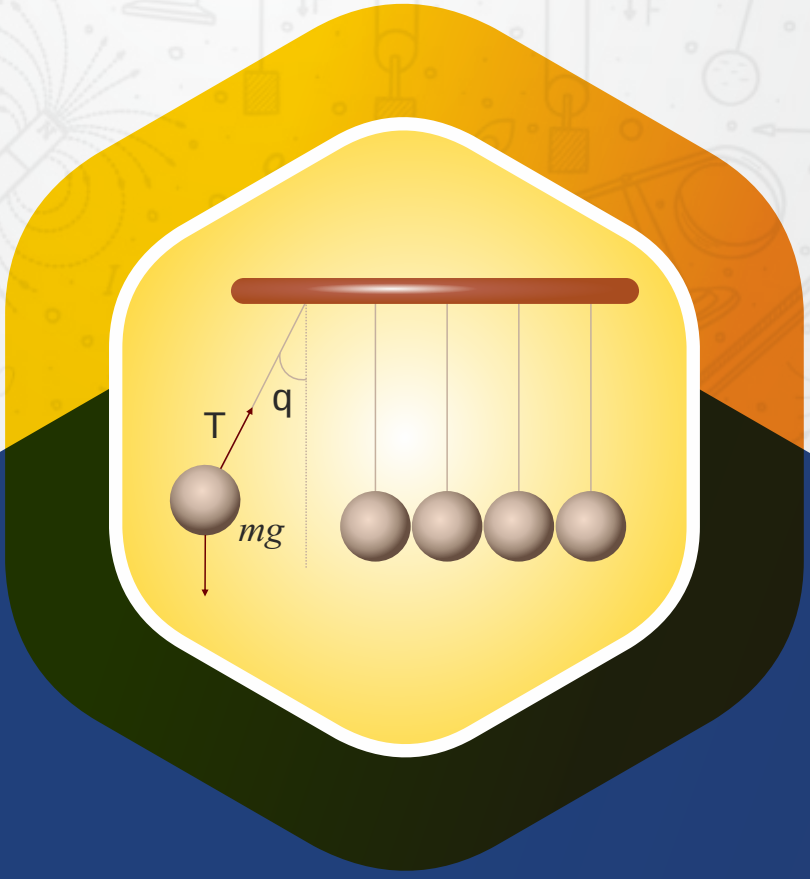
ALLEN CAREER INSTITUTE
KOTA (RAJASTHAN)
path to success

PHYSICS

JEE (MAIN+ADVANCED)

NURTURE COURSE



Study Material

Newton's Law of Motion & Friction

(English Medium)

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NEWTON'S LAWS OF MOTION & FRICTION

KEY CONCEPT

FORCE

The concept of force is used to explain mutual interaction between two material bodies as the action of one body on another in form of push or pull, which brings out or tries to bring out a change in the state of motion of the two bodies.

GRAVITATIONAL FORCE

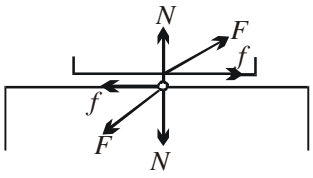
It is a force of pull of Earth on a body towards the centre of Earth i.e. in vertical direction. It is a distributed force but the net force can be assumed to be acting on centre of gravity of body. For bodies of negligible size in comparison to Earth, it can be assumed to be acting at the centre of mass of the body.

CONTACT FORCE

When two bodies come in contact they exert forces on each other. These forces are called contact forces. There are two components of contact force.

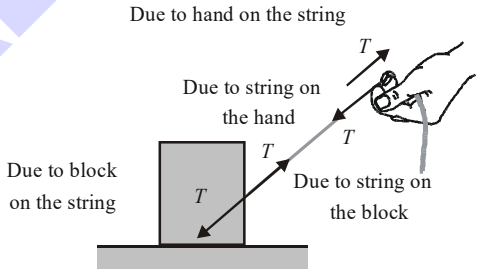
(i) **Normal force (*N*) :** It is the component of contact force normal to the surface. It measures how strongly the surfaces in contact push each other.

(ii) **Frictional force (*f*) :**
It is the component of contact force parallel to the surface.
It opposes the relative motion (or attempted relative motion) of the two surfaces in contact.



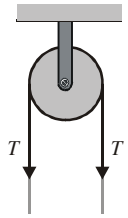
TENSION FORCE (*T*)

The force of tension at a section of a string is the force of pull applied by one part of a taut string, rope or chain to its other part through that section, along its length. It is also the force applied by taut string or chain at its end to the objects connected to it, along its length.



STRING PASSING OVER A PULLEY

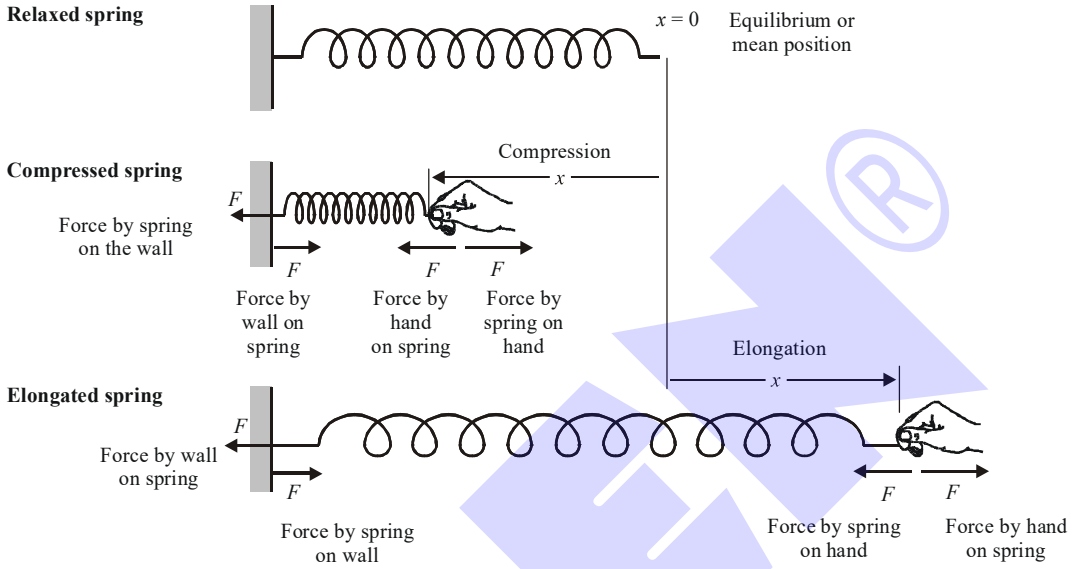
A pulley is a device consisting of a wheel, which can rotate freely on its axel. A single pulley changes direction of tension force. At present for simplicity, we discuss only ideal pulley, which is massless i.e. has negligible mass and rotates on its axel without any friction. An ideal pulley offers no resistance to its rotation, therefore tension force in the string on both sides of it are equal in magnitude. Such a pulley is known as ideal pulley.



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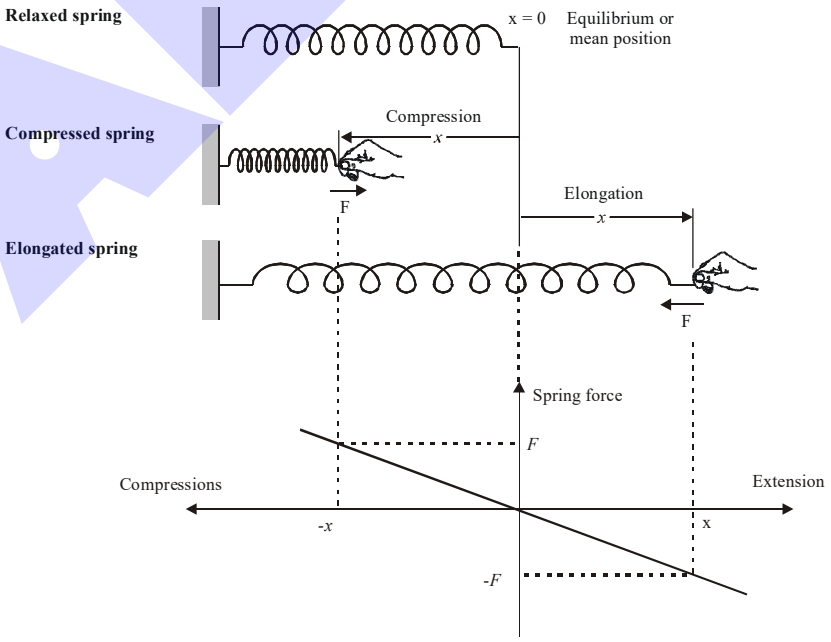
SPRING FORCE

When no force acts on a spring, it is in relaxed condition i.e. neither compressed nor elongated. Consider a spring attached to a fixed support at one of its end and the other end is free. If we neglect gravity, it remains in relaxed state. When it is pushed by a force F, it is compressed and displacement x of its free end is called compression. When the spring is pulled by a force F, it is elongated and displacement x of its free end is called elongation. Various forces developed in these situations are shown in the following figure.



The force applied by the spring on the wall and the force applied by the wall on the spring make a third law action-reaction pair. Similarly, force by hand on the spring and the force by spring on the hand make another third law action-reaction pair.

Hooke's Law: How spring force varies with deformation in length x of the spring is also shown in the following figure.



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The force F varies linearly with x and acts in a direction opposite to x. Therefore, it is expressed by the following equation $F = -kx$

Here, the minus (–) sign represents the fact that force F is always opposite to x.

The constant of proportionality k is known as *force constant of the spring* or simply as *spring constant.*

The slope modulus of the graph equals to the spring constant.

SI unit of spring constant is newton per meter or (N/m).

Dimensions of spring constant are MT^{-2}.

NEWTON'S LAWS OF MOTION

The First Law

Every material body has tendency to preserve its state of rest, or of uniform motion in a straight line, unless it is compelled to change that state by external forces impressed on it.

- **Inertia**

  The tendency of a material body to preserve its present state of uniform motion or of rest is known as inertia of the body. It was first conceived by Galileo.

  Inertia is a physical quantity and mass of a material body is measure of its inertia.

- **Inertial Frame of Reference**

  The first law requires a frame of reference in which only the forces acting on a body can be responsible for any acceleration produced in the body and not the acceleration of the frame of reference. These frames of reference are known as inertial frames.

The Second Law

The rate of change in momentum of a body is equal to, and occurs in the direction of the net applied force. A body of mass m in translational motion with velocity $\vec{v}$, if acted upon with a net external force $\vec{F}$, the second law suggests:

$$\vec{F} = \frac{d}{dt}(m\vec{v})$$

If mass of the body is constant, the above equation relates the acceleration $\vec{a}$ of the body with the net force $\vec{F}$ acting on it.

$$\vec{F} = \frac{d}{dt}(m\vec{v}) = m\vec{a}$$

The first law provides concept of force and the second law provides the quantitative definition of force, therefore the second law is also valid only in inertial frames.

SI unit of force is newton. It is abbreviated as N. One newton equals to one kilogram-meter per second square.

$$1\ \text{N} = 1\ \text{kg-m/s}^2$$

Dimensions of force are MLT^{-2}

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The Third Law

Force is always a two-body interaction. The first law describes qualitatively and the second law describes quantitatively what happens to a body if a force acts on it, but do not reveal anything about what happens to the other body participating in the interaction responsible for the force.

The third law accounts for this aspect of the force and states that every action on a body has equal and opposite reaction on the other body participating in the interaction.

CONCEPT OF FREE BODY DIAGRAM (FBD)

A free body diagram is a pictorial representation in which the body under study is assumed free from rest of the system i.e. assumed separated from rest of the interacting bodies and is drawn in its actual shape and orientation and all the forces acting on the body are shown.

How to draw a Free Body Diagram (FBD)

- Separate the body under consideration from the rest of the system and draw it separately in actual shape and orientation.
- Show all the forces whether known or unknown acting on the body at their respective points of application.

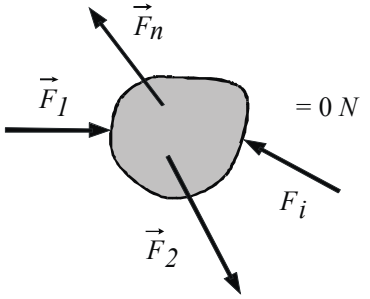
  For the purpose count every contact where we separate the body under study from other bodies. At every such point, there may be a contact force. After showing, all the contact forces show all the field forces.

TRANSLATIONAL EQUILIBRIUM

A body in state of rest or moving with constant velocity is said to be in translational equilibrium. Thus if a body is in translational equilibrium in a particular inertial frame of reference, it must have no linear acceleration. When it is at rest, it is in *static equilibrium*, whereas if it is moving at constant velocity it is in *dynamic equilibrium*.

Conditions for translational equilibrium

For a body to be in translational equilibrium, no net force must act on it i.e. vector sum of all the forces acting on it must be zero.



If several external forces $\vec{F}_1$, $\vec{F}_2$ ..... $\vec{F}_i$ ..... and $\vec{F}_n$ act simultaneously on a body and the body is in translational equilibrium, the resultant of these forces must be zero.

$$\sum \vec{F}_i = \vec{0}$$

If the forces $\vec{F}_1$, $\vec{F}_2$ ..... $\vec{F}_i$ ..... and $\vec{F}_n$ are expressed in Cartesian components,

we have : $\sum F_{ix} = 0$ $\sum F_{iy} = 0$ $\sum F_{iz} = 0$

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PSEUDO FORCE

The magnitude of this force F_P is equal to the product of the mass m of the object and acceleration a of the frame of reference. The direction of the force is opposite to the direction of acceleration.

$$F_P = -ma$$

It is an imaginary force and assumed to be acting on a body when we want to apply NLM in a non-inertial frame.

FORCE OF FRICTION

Comes into action only when there is a slipping between the two contact surfaces or when an attempt is made to have it.

The force of friction on each body is in a direction opposite to its motion (existing or impending) relative to other body.

(a) **Static friction :** The frictional force acting between any two surfaces at rest with respect to each other is called the force of static friction (f_s).

$$f_s \le \mu_s N$$

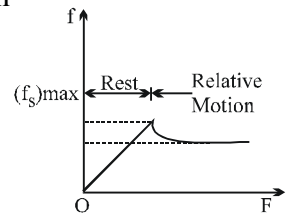
where μ_s is the coefficient of static friction.]

(b) **Kinetic friction :** The frictional force acting between surfaces slipping on each other is called the force of kinetic friction or sliding friction (f_k).

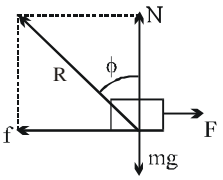
$$f_k = \mu_k N$$

where μ_k is the coefficient of kinetic friction.

$$\mu_s > \mu_k$$



Angle of friction (ϕ) : Mathematically, the angle of friction (ϕ) may be defined as the angle between the normal reaction N and the total contact force R.



Thus $\tan\phi = \dfrac{f}{N}$

Since $f = \mu N$, therefore,

$$\tan\phi = \mu$$

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